Worksheet — Impulse Simulation

Use with *Simulation - Impulse.html* · momentum and impulse (Block 6): Smart Cart and force sensor, egg into a sheet, ball, racket and club, safety (airbags, seat belts, helmets, crumple zones, landing)

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Name: _____ Class: _____ Date: _____

How to use this worksheet

Scan the QR code or open **school.taneski.com/physics/dp/17-impulse** in a browser (offline: *Simulation - Impulse.html*). Choose an example at the top: **Cart** + force sensor · **Egg** into a sheet · **Ball**, racket, club · **Safety** (airbags, seat belts, helmets, crumple zones, landing). Set values with ▼ ▲, or type a value in the box.

Choose the **positive direction** (+ → right or + ← left) in the controls: every velocity, momentum, force and impulse takes its sign from it. The contact plays in slow motion; ◂ / ▸ step through it. On the *F*–*t* graph the **Area** tool gives the impulse; tick **Zoom on the contact**.

Part 1 before the lab (≈ 15 min) · **Part 2** during the lab (≈ 20 min) · **Part 3** homework (≈ 30 min) · **Part 4** Safety (≈ 30 min, class or homework). The Tasks tab has the same three parts. Use $p = mv$, $J = F\Delta t = \Delta p$. Give answers to 3 significant figures with units.

Scan to open

Part 1 — Pre-lab: predict, then test

Question	My prediction (before)	What the simulation showed	Physics explanation
1. 🛒 Cart, 0.500 kg at 0.62 m s^{-1}. Magnetic bumper instead of spring bumper: is the **peak force** larger, smaller or the same?			
2. Is the **impulse** (area under *F*–*t*) for the magnetic bumper much larger, much smaller or about the same?			
3. With **clay** on the bumper the cart stops at the sensor. Its impulse is about × ____ the spring bumper's.			
4. 🥚 An egg is thrown at 10 m s^{-1} into a brick wall or into a loosely held bedsheet. Which is the same: Δp, Δt or F? Which is different?			

5. 🛒 Spring bumper: the cart (0.500 kg) arrives at +0.62 m s^{-1} and leaves at −0.496 m s^{-1} (+ is to the right). Calculate the change in momentum Δ*p*, with its sign. Then check the readouts.

Working

6. 🥚 Calculate the average force of the sheet on a 60 g egg that arrives at 10 m s^{-1} and is stopped in 0.10 s.

Working

7. Sketch, on the same axes, the force–time graphs you expect for the spring bumper and for the magnetic bumper. Label the peak forces and contact times you read from the simulation. Shade the impulse of one of them.

Sketch: force F / N against time t / s

Part 2 — During the lab

A · Smart Cart and force sensor (PASCO demo). Simulation: 🛒 Cart, m = 0.500 kg, u = 0.62 m s^{-1}, + → right. For each bumper press ▶️ Start, tick **Zoom** and **Area** on the F–t graph, step to the end of the contact and press ● Record. Write the class Capstone values in the last column.

bumper	contact time Δ*t* / s	peak force / N	impulse *J* (area) / N s	*v* after / m s^{-1}	Δ*p* = *m*(*v* − *u*) / N s	class Capstone *J* / N s
spring						
magnetic						
clay						

8. Compare the impulse *J* with Δ*p* for each bumper. What do you notice? Calculate the percentage difference between the simulation and the class value for the spring bumper.

9. Explain why the impulse with clay is only about half of the impulse with the spring bumper, although the peak forces are similar.

B · Faster carts. Spring bumper, m = 0.500 kg. Record the peak force at each speed.

u / m s^{-1}	0.31	0.62	0.93	1.24
peak force / N				
contact time Δt / s				

10. Describe how the peak force and the contact time depend on the speed. Use the Data tab (F_{peak}–u) to support your answer.

Part 3 — Homework

11. 🏌 (slide 5) A golf club hits a 0.046 kg ball at rest. The force rises steadily to 800 N and falls back to zero; the contact lasts 6.0 ms (a triangle on the *F*–*t* graph). Calculate the impulse and the speed of the ball. Check: Ball → Golf ball + club.

Working

12. 🎾 (slide 8) A 0.058 kg tennis ball arrives at 20 m s^{-1} and leaves at 30 m s^{-1} in the opposite direction; contact 5.0 ms. Take + as the outgoing direction. Calculate Δ*p* and the average force of the racket on the ball.

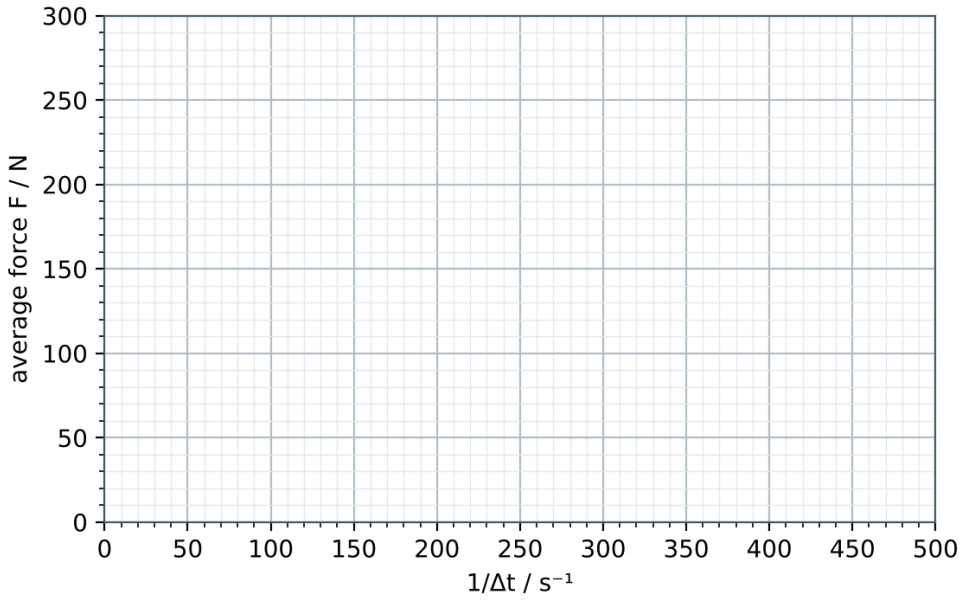
Working

13. A 0.40 kg ball hits a wall at 10 m s^{-1} and rebounds at 8.0 m s^{-1}; contact 0.050 s. Calculate the size of the average force. Why is it larger than if the ball had stopped dead?

Working

14. 🥚 Egg, 60 g at 10 m s^{-1}. Type each stopping time into the Δ*t* box and record the average force (Data tab).

Δt / ms	2	5	10	20	50	100
1/Δt / s^{-1}						
average force / N						
egg cracks? (peak > 25 N)						



Plot the average force against 1/Δ*t*, draw the best-fit line through the origin and find its gradient. Gradient = _____ . What does it represent?

15. In this model the peak force is π/2 × the average force. Calculate the shortest stopping time for which the egg survives (peak force below 25 N). Check your answer in the simulation.

Working

16. Use the idea of impulse to explain how a car's crumple zone and airbag protect the driver.

Part 4 — Safety: the same impulse, a smaller force

Simulation: 🚗 Safety. Choose a situation. Every lane stops the same body (same *m* and *u*) but over a different stopping time Δ*t*. Tap a lane to see its forces; Compare (Graphs tab) shows Δ*p* and the forces as bars. + is the direction of the motion.

A · Predict (before you press ▶️ Start).

Question	My prediction (before)	What the simulation showed	Physics explanation
17. 🚗 Car driver, 70 kg at 15 m s^{-1}: ① no seat belt, ② seat belt, ③ seat belt + airbag. Which lane has the **largest impulse** on the driver?			
18. Which lane has the **smallest average force**? Which lane(s) give a serious injury (peak above 30 *g*)?			

B · During the lab.

19. For each situation press ▶️ Start and read the table under the scene (①, ②, ③ = the lanes, from the top or from the left). First calculate Δ*p* = 0 − *mu* yourself for one lane of each situation, then complete the table.

situation · lane	Δ*t* / s	Δ*p* / N s	F_{avg} / N	F_{peak} / N	injury limit / N	serious injury?
Driver ①						
Driver ②						
Driver ③						
Head ①						
Head ②						
Car ①						
Car ②						
Landing ①						
Landing ②						
Landing ③						

20. 🚗 Car body (as on the slides): a 1200 kg car at 15 m s^{-1} hits a wall. Calculate the average force on the car for the rigid car (Δt = 0.050 s) and with a crumple zone (Δt = 0.25 s). How many times smaller is the force with the crumple zone?

Working

21. Open Compare in the Graphs tab. Describe what the two sets of bars show, and use $F = \Delta p/\Delta t$ to explain why one set is equal and the other is not.

C · Homework.

22. 🦘 Landing (backup slide 21): a 60 kg student lands at 4.0 m s^{-1}. Calculate the average force of the floor with stiff legs (Δt = 0.020 s) and with bent knees (Δt = 0.40 s). Give each force as a multiple of the student's weight (g = 9.81 N kg^{-1}). Ignore the weight during the stop, as the simulation does.

Working

23. 🚲 A cyclist's head (5.0 kg) hits the road at 6.0 m s^{-1}. A serious head injury starts at a peak deceleration of 300 g. In this model $F_{peak} = (\pi/2) \times F_{avg}$. Calculate the shortest stopping time the helmet foam must give. Check it with the Δt ② box.

Working

24. 🚗 Car driver (70 kg, 15 m s^{-1}). Type the stopping times below into the Δ*t* boxes (three at a time) and ● Record. Complete the table.

Δ*t* / ms	10	20	50	100	150	300
$1/\Delta t$ / s^{-1}						
average force / N						
serious injury? (peak > 20.6 kN)						

Plot the average force against $1/\Delta t$ (graph paper or the Data tab → F_{avg}–$1/\Delta t$), draw the best-fit line through the origin and find its gradient. Gradient = _____ . What does it represent?

25. A student writes: "An airbag protects the driver because it reduces the impulse on the driver." Explain what is wrong and write a correct statement.